Munir Awwad

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Highlights of Qualifications

- Enrolled in the fourth year of the Computer Science CO-OP (Honours) program at McMaster University.
- Programming experience in Python, Java, C#, C, HTML, CSS, JavaScript/TypeScript, and React.
- Experience with version control, CI/CD pipelines, databases, Linux systems, and networking.
- Familiar with the software development life cycle and Agile development principles, including Scrum and XP.
- Self-learner with an enthusiasm for acquiring new skills, exhibited through self-studying web and app development.

Work Experience

Software Analyst Intern

Sep 2025 - Aug 2026 (12 months)

Hitachi Rail - Toronto, ON, Canada

- Designed and built a scalable Playwright-based automated regression testing framework from the ground up alongside a fellow intern, enabling automated front-end testing of an automatic train supervision (ATS) system.
- Developed and maintained 75+ automated test cases covering both application functionality and business requirements, improving test coverage and reliability.
- Integrated the testing framework with Jenkins to automatically build the software, execute test suites, and generate a detailed test report with requirements traceability.
- Contributed directly to the ATS software by fixing defects and supporting migration from older dependencies to newer ones, while gaining hands-on experience with JavaScript libraries such as PixiJS and StencilJS.

Education

B.A.Sc. Computer Science CO-OP (Honours)

Sep 2022 - Apr 2027

McMaster University, Hamilton ON

- **GPA**: 3.79/4.00 (Dean's Honour List)
- **Highlights**: Data Structures, Algorithms and Complexity, Object-Oriented Programming, Databases, Networks and Security, Numerical Methods, Machine Learning, Natural Language Processing, Applied Cryptography.

Professional Development

- IBM Application Security for Developers and DevOps Professionals – Coursera Certificate (In Progress)
- Playwright: Web Automation Testing From Zero to Hero – Udemy Certificate
- JavaScript Unit Testing - The Practical Guide – Udemy Certificate

Technical Skills

- **Languages**: Python, Java, C, JavaScript/TypeScript, HTML, CSS, SQL, C#, Haskell, Bash, Julia, PHP, R.
- **Frameworks and Libraries**: ReactJS, React Native, Playwright, Jest, Vitest, PyTorch, Scikit-learn, Gensim, NLTK.
- **Version Control and CI/CD**: Git, GitHub, Bitbucket, Jenkins, GitHub Actions.
- **Databases**: SQL, IBM Db2, ER diagrams, schema design, schema normalization / decomposition, execution plans.
- **Linux Systems**: Multiprocessing, multithreading, GCC, Bash scripting, Makefiles, kernel module development.
- **Networking**: OSI model, TCP/IP model, socket programming, Cisco Packet Tracer.
- **Software Development**: SDLC, UML, Agile development, Scrum, XP, plan-driven development, Waterfall.
- **Additional Tools and Technologies**: NodeJS, npm, Jira, Jupyter (IPython) Notebooks.

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Projects and Coursework

- **AI Gatekeeper for Biomedical and Scientific Journals** (Capstone Project – In progress):
  - Contributing to the development of an AI-powered manuscript screening application, including transitioning to locally hosted models with expanded capabilities for multimodal analysis and detection of research integrity violations.
  - Leading security and reliability efforts, including assessing and mitigating common web application vulnerabilities (OWASP Top 10), protecting against prompt injection attacks, and evaluating secure data handling practices.
  - Reviewing encryption, hashing, and other security mechanisms to ensure data confidentiality, integrity, and application reliability.
- **Digit Classification and Image Denoising with Convolutional Neural Networks** (University Coursework):
  - Constructed and trained CNNs that leverage kernels and pooling to classify images of single digit building numbers into the classes 0-9, as well as CNNs that denoise these images when Gaussian noise is applied.
  - Implemented in Python using the PyTorch library, achieving an accuracy score of 92% on a test set of 26,000 images for classification, and a mean squared error loss of < 0.01 on a test set of 15,000 images for denoising.
- **SMS Spam Detection by LLM Fine-Tuning** (University Coursework)**:**
  - Preprocessed and tokenized a text dataset of SMS messages and used them to fine-tune variations of the BERT and T5 LLMs to the task of spam detection.
  - Code written in Python, making use of NumPy, Scikit-Learn, NLTK, and Gensim, achieving a test accuracy score of 99% on a test set of 560 messages.
- **Weather Application** (Personal Project):
  - Developed a weather app for Android that allows users to enter city names and utilizes asynchronous API calls to fetch and dynamically display relevant weather data through the app's user interface.
  - Built using React Native and the Expo framework.
- **To-Do List Chrome Extension** (Personal Project)**:**
  - Developed a Google Chrome extension to help users manage tasks and task progress from within the browser.
  - Created using HTML, CSS, and JavaScript.